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Programme-B. Tech CSE (AI&ML)

Section-F

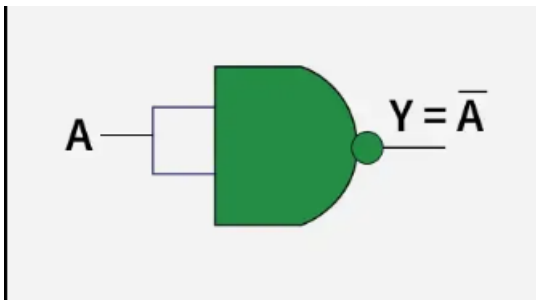
Subject Name-Nand To Tetris-1

ASSESSMENT-1

QUE-1 Implement NOT ,AND ,OR gets using NAND gates in HDL .Draw the logic diagram for each and write the Boolean expression. Run the provided test scripts and capture the simulation output.

NOT GATE-

Logic Diagram-



NOT Gate Truth Table-

A	Y
0	1
1	0

Boolean Expression-

$$Y = A \text{ NAND } A = \text{NOT } A$$

HDL Dashboard-

HDL
--LBSt
Project 1
Not

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.hdl
5 /**
6  * Not gate:
7  * if (in) out = 0, else out = 1
8  */
9 CHIP Not {
10     IN in;
11     OUT out;
12
13     PARTS:
14     Nand(a=in , b= in, out=out );
15 }

```

Simulation successful: The output file is identical to the c

Test Script-

Test Script	Compare File	Output File	Diff Table
<pre> 1 This file is part of www.nand2tetris.org 2 and the book "The Elements of Computing Systems" 3 by Nisan and Schocken, MIT Press. 4 File name: projects/1/Not.tst 5 6 d Not.hdl, 7 pare-to Not.cmp, 8 put-list in out; 9 10 in 0, 11 l, 12 put; 13 14 in 1, 15 l, 16 put; </pre>			

Chip Not-

Chip Not

Eval Reset Clock: 0

Input pins

in 1

Output pins

out 0

compare file

Compare File-

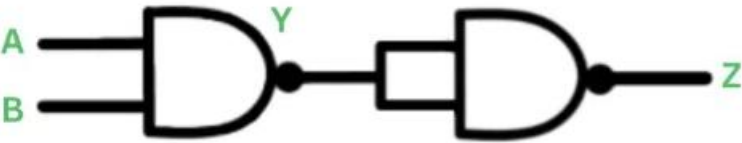
|in |out|

| 0 | 1 |

| 1 | 0 |

AND GATE-

Logic Diagram-



AND Gate Truth Table-

A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

Boolean Expressions-

$$X = A \text{ NAND } B = (A \cdot B)'$$

$$Z = X \text{ NAND } X = A \cdot B$$

HDL Dashboard-

The screenshot shows the HDL Dashboard interface. At the top, there's a header with 'HDL' and a toggle switch, followed by 'Project 1' and 'And'. Below this is a code editor with the following Verilog code:

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.hdl
5 /**
6  * And gate:
7  * if (a and b) out = 1, else out = 0
8  */
9 CHIP And {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         Nand(a=a,b=b,out=nandOut);
15     Not(in=nandOut,out=out);
16
17
18 }

```

At the bottom of the dashboard, a green status bar displays the message: "Simulation successful: The output file is identical to the c".

Test Script-

Test**Steps: 7 Outputs: 5**

Slow



Fast

Test Script**Compare File****Output File****Diff Table**

```
2 and the book "The Elements of Computing Systems"
3 by Nisan and Schocken, MIT Press.
4 File name: projects/1/And.tst
5
6 d And.hdl,
7 pare-to And.cmp,
8 put-list a b out;
9
10 a 0,
11 b 0,
12 l,
13 put;
14
15 a 0,
16 b 1,
17 l,
18 put;
19
20 a 1,
21 b 0,
22 l,
23 put;
24
25 a 1,
26 b 1,
27 l,
28 put;
```

Chip And-

Chip And

Eval Reset Clock: 0

Input pins

a

1

b

1

Output pins

out

1

Internal pins

nandOut

0

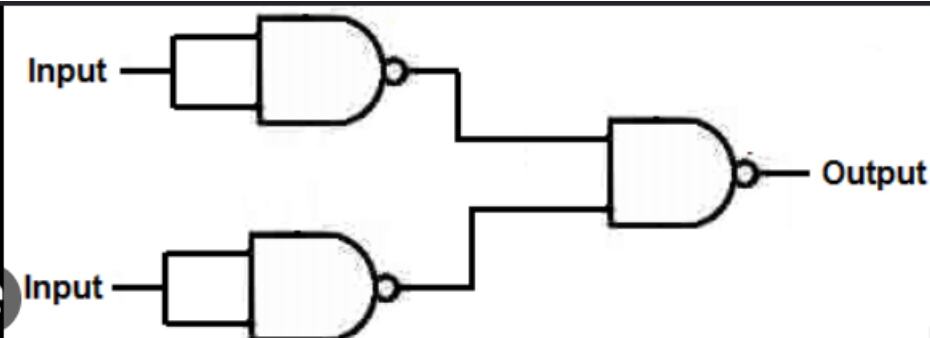
compare file

Compare File-

a b out
0 0 0
0 1 0
1 0 0
1 1 1

OR GATE-

Logic Diagram-



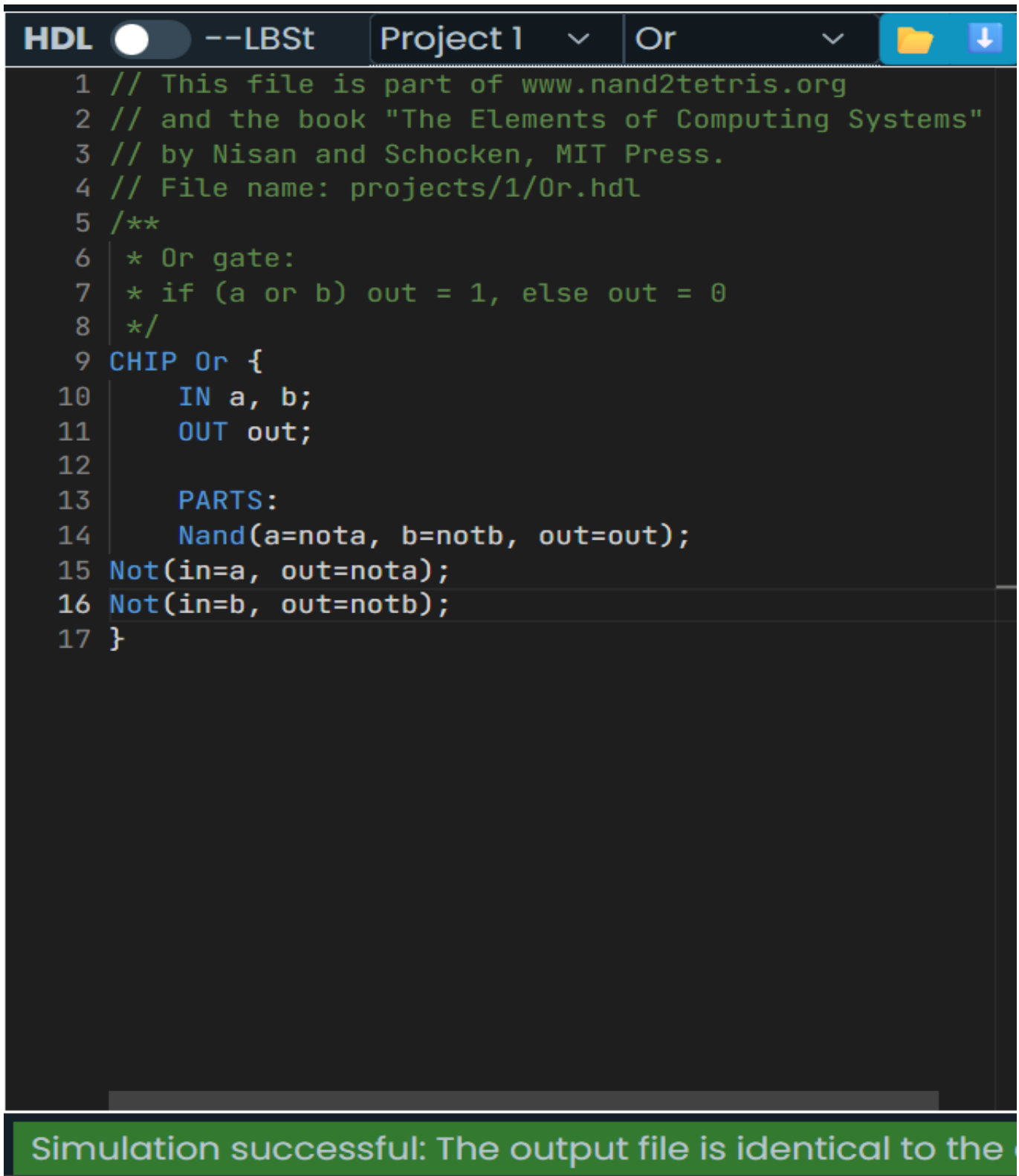
OR Gate Truth Table-

A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

Boolean Expressions-

$A' = A \text{ NAND } A$
 $B' = B \text{ NAND } B$
 $Y = A' \text{ NAND } B' = A + B$

HDL Dashboard-



```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or.hdl
5 /**
6  * Or gate:
7  * if (a or b) out = 1, else out = 0
8  */
9 CHIP Or {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         Nand(a=nota, b=notb, out=out);
15     Not(in=a, out=nota);
16     Not(in=b, out=notb);
17 }
```

Simulation successful: The output file is identical to the

Test Script-

Test

Slow

Fast

Steps: 1 Outputs: 5

Test Script

Compare File

Output File

Diff Table

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

and the book The Elements of Computing Systems

by Nisan and Schocken, MIT Press.

File name: projects/1/Or.tst

d Or.hdl,

pare-to Or.cmp,

put-list a b out;

a 0,

b 0,

l,

put;

a 0,

b 1,

l,

put;

a 1,

b 0,

l,

put;

a 1,

b 1,

l,

put;

Chip Or-

Chip Or

Eval

Reset

Clock: 0

Input pins

a

1

b

1

Output pins

out

1

Internal pins

nota

0

notb

0

compare file

Compare File-

| a | b | out |

| 0 | 0 | 0 |

| 0 | 1 | 1 |

| 1 | 0 | 1 |

| 1 | 1 | 1 |

